



Dental Cements (Part -2)



Learning outcomes

- Recognize the importance of pulp protection
- Summarize about cavity varnish, liners and bases
- Explain Glass Ionomer cement composition, setting reaction, manipulation, properties and indications.

INTRODUCTION:

The goals of restorative therapy are not only to restore the tooth to proper form and function, but also to minimize postoperative sensitivity and preserve pulp vitality.

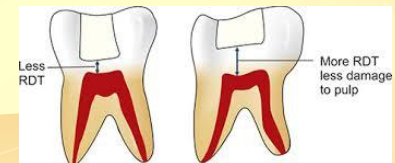
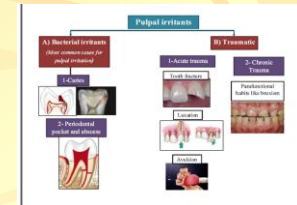
Main irritants are

➤ Caries(Most common)

➤ Cavity preparation procedure

Restorative materials used

Microleakage



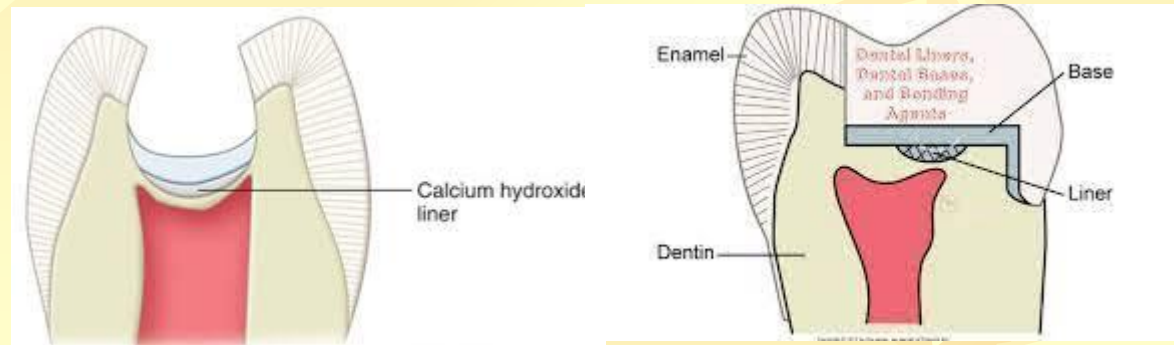
IDEAL REQUIREMENTS OF PULP PROTECTING AGENTS

- BICOMPATIBLE
- SET/HARDEN QUICK
- LOW SOLUBILITY IN SALIVA
- SHOULD WITHSTAND THE CONDENSATION FORCES
- STABILITY AND FURTHER DECREASE DENTIN PERMEABILITY
- EASE OF USE AND MANIPULATION

Cavity Liner

■ *Definition by Journal of Operative Dentistry*

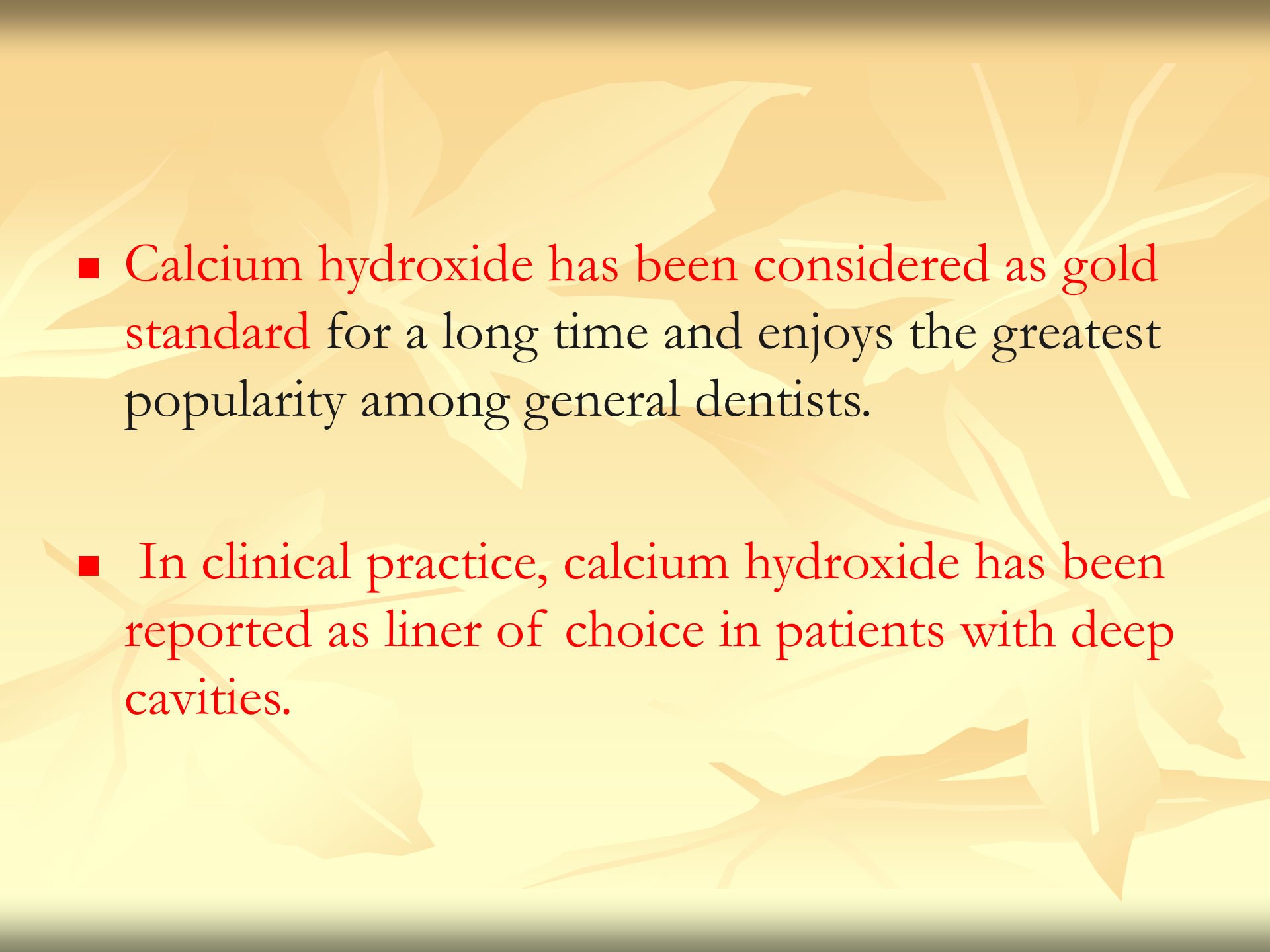
Cavity liner is a dental cement or resin coating with minimal thickness (usually less than 0.5 mm) to achieve a physical barrier to bacteria or use as therapeutic effect



- A variety of dental materials have been introduced as liners to provide pulp tissue protection from irritants related to the restorative procedure.
- The traditional lining materials are
 - Calcium hydroxide
 - Glass ionomer
 - Resin-modified glass ionomers (RMGIs)

Liners and bases in general dentistry

R Weiner, Australian Dental journal

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- Calcium hydroxide has been considered as gold standard for a long time and enjoys the greatest popularity among general dentists.
 - In clinical practice, calcium hydroxide has been reported as liner of choice in patients with deep cavities.

Calcium Hydroxide

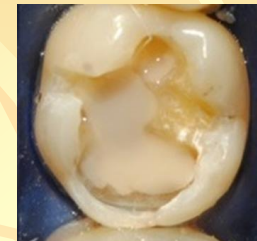
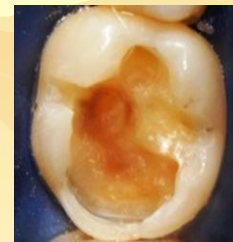
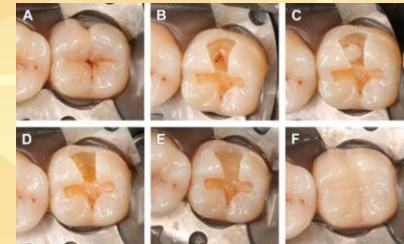
- First used by Herman in 1930.

Actions of calcium hydroxide - The hydroxyl ions create an alkaline pH that is unfavorable for remaining bacteria in the cavity.



Indications:

- For Direct pulp capping
- For indirect pulp capping



Modes of supply:

- Two paste system- base/catalyst paste in collapsible tubes.



Composition- Dycal

- Base paste
 - Glycol Salicylate – 40 %
 - Calcium sulphate
 - Titanium dioxide
 - Barium sulphate : provides radiopacity

■ Catalyst paste

- CaOH : 50% : principal reactor
- ZnO 10%
- Zn stearate 0.5%
- Ethylene toluene
- Sulfonamide 39.5 % : oily compound act as carrier

■ Setting reaction



Mixing time: 10 seconds

Setting time :-- 2.5-3.5 min

Properties

1. Mechanical

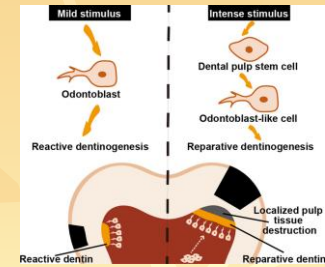
- Compressive strength: Low but increases with time after setting after 24 hrs 10-27 MPa
- Tensile strength : Low (1 MPa)

2. Biologic properties

- Effect on pulp : alkaline ; high pH pH range is 9.2-11.7
- Formation of reparative dentin.

Advantages:

1. Stimulates reparative dentin formation
2. Biocompatible
3. High pH



Disadvantages:

1. Low strength
2. High solubility
3. Used over small areas requiring pulp protection - Applying glass ionomer base over calcium hydroxide to prevent its dissolution.



■ Mode of supply

Self cured

- Dycal

Light cure

- Prisma VLC Dycal

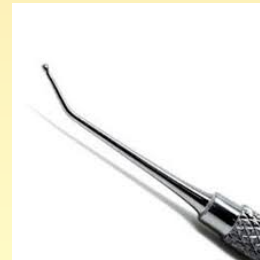


➤ Manipulation technique

Dispense equal volumes of base and catalyst pastes on the paper pad

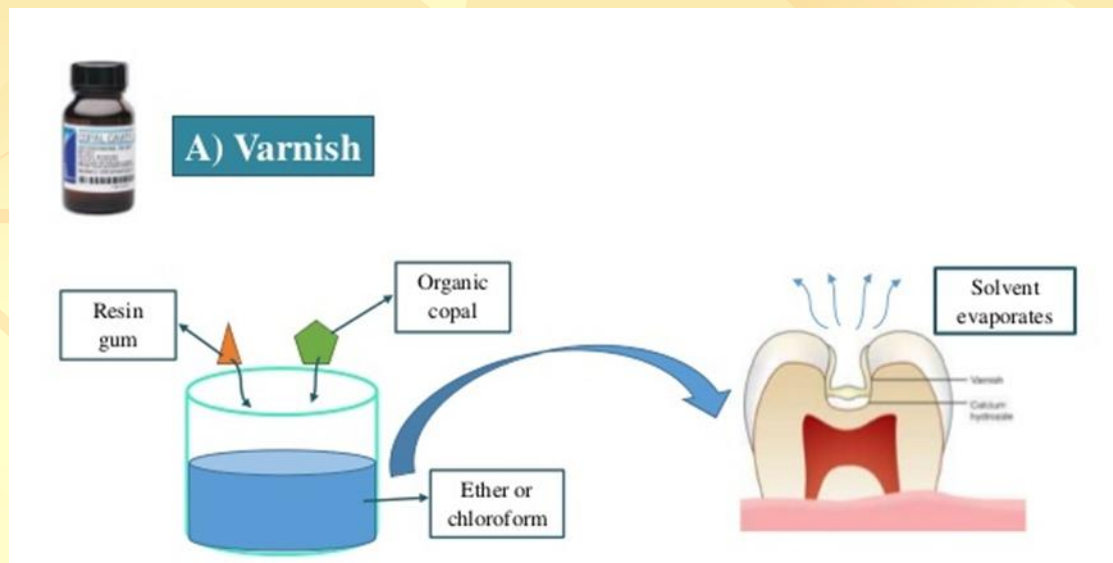
Using a Dycal Liner applicator, mix thoroughly until a uniform color is achieved

Complete mixing within 10 seconds.



Cavity Varnish

- Cavity varnish is a solution of one or more resins which when applied onto the cavity walls, evaporates leaving a thin resin film, that serves as a barrier between the restoration and the dentinal tubules.



- ▶ Two coats of varnish should be applied using a disposable microapplicator to ensure sufficient wetting of cavity walls (One application will give voids – so apply 2 times)
- ▶ Varnish has to be applied to all cavity walls and pulpal floor



Indications for cavity varnish:

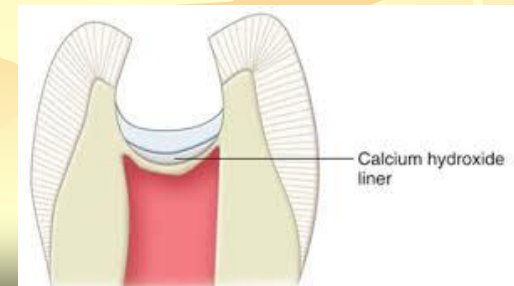
1. Below amalgam restoration to reduce microleakage
2. Below Zinc Phosphate Cement

Contraindication for cavity varnish

1. Below composite restorations, interferes with polymerization of resin

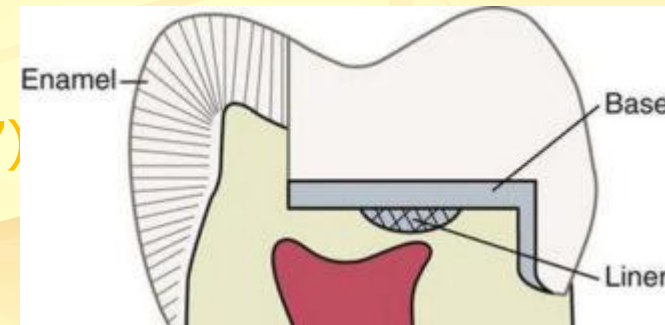
Cavity base

- Is a layer of cement that is placed under the permanent restoration to protect the pulp against numerous types of irritation is called cavity Base.
- Marzouk defined cavity bases as insulating materials that can be used directly on certain areas of the dentinal parts of the preparation.



- Base is a replacement for protective dentin, which has been destroyed during caries process or during cavity preparation or both.
- Base is typically 0.5 -1mm thickness

(STURDEVANT-6th Edition, Chapter 5-PAGE-157)



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- Bases are applied in thick layers to provide the pulp with thermal protection.
 - These materials must be strong enough to support a restorative material during placement and function.

- Craig and Powers subdivided bases into low strength and high strength categories
- High-strength bases used to provide mechanical support for a restoration and thermal protection for the pulp. These include zinc phosphate, zinc polycarboxylate, glass ionomers and polymer-reinforced zinc oxide eugenol.

Glass Ionomer cement

- Glass-ionomer (GIC) material was introduced by Wilson and Kent in 1972.
- It was initially introduced as a “new dental filling material” recommended for the restoration of cervical lesions.

- The glass - ionomer family of restorative materials has evolved during the past 40 years into a diverse use that includes direct restoratives, luting agents, liners, and bases as well as pit and fissure sealants, all available in both the conventional and resin - modified varieties

Classification

■ ACCORDING TO SKINNERS

- Type I - Luting
 - Type II- Restorative : Available as Conventional as well as resin modified
 - Type III- Liners and bases
- ## ■ Metal modified Glass Ionomer
- a. Miracle mix
 - b. Cermet cement

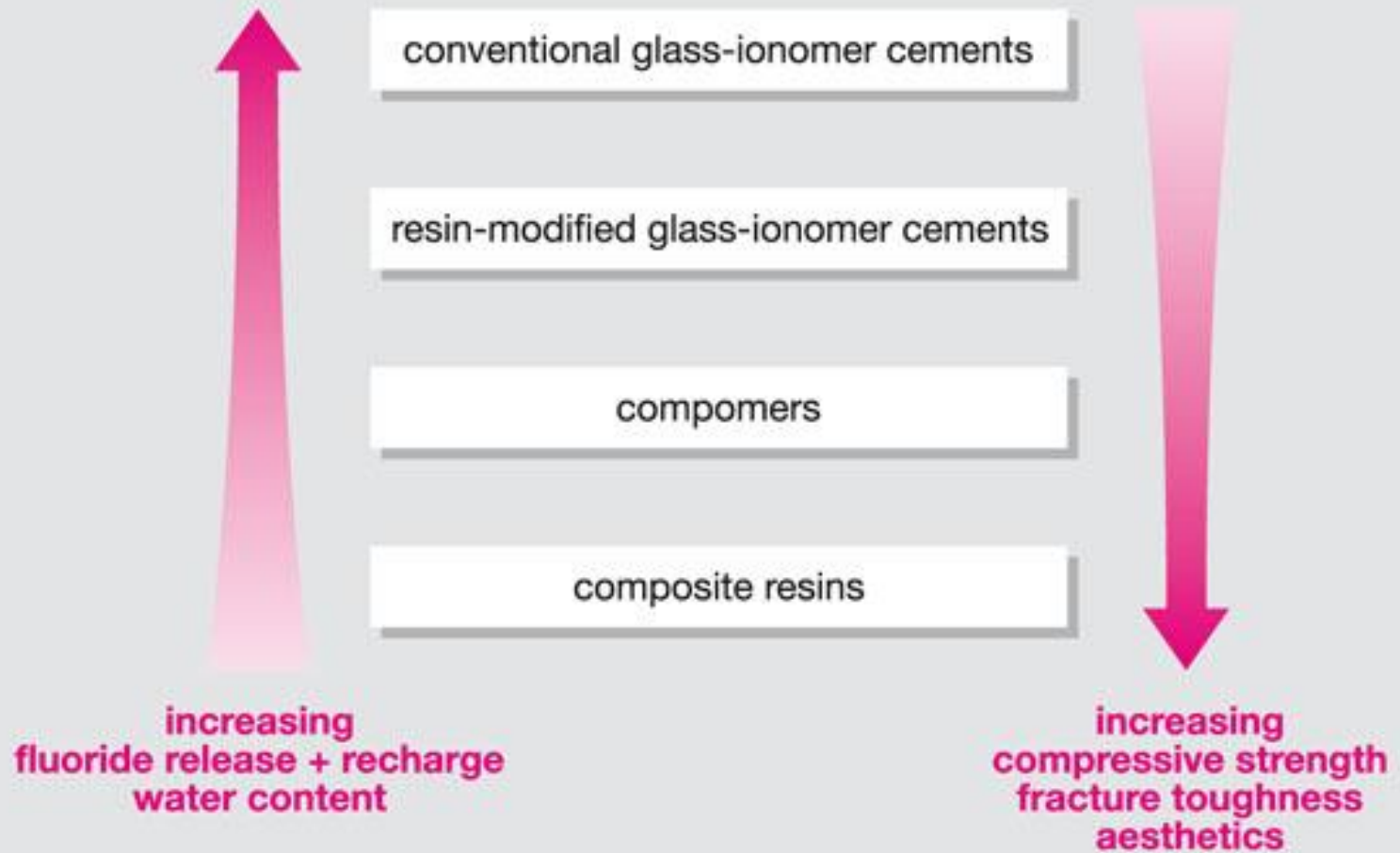


■ Hybrid Glass Ionomer

➤ Compomers

➤ Giomers





Composition

POWDER

- Silica (SiO_2) - 29%
 - Alumina (Al_2O_3) - 16.6%
 - Aluminium Fluoride (AlF_3) - 5.3%
 - Calcium Fluoride (CaF_2) - 34.2%
 - Sodium Aluminum Fluoride (Na_3AlF_6) - 5%
 - Aluminium phosphate (AlPO_4) - 9.9%
 - Lanthum ; Strontium ; Barium ; radioopacity
- It basically consists of fluoroaluminosilicate glass powder

LIQUID

- Polyacrylic acid -40% - Formation of cement matrix
- Water -50%
- Itaconic acid -5% - Decrease viscosity of liquid
- Tartaric acid -5-15% - Increases working time

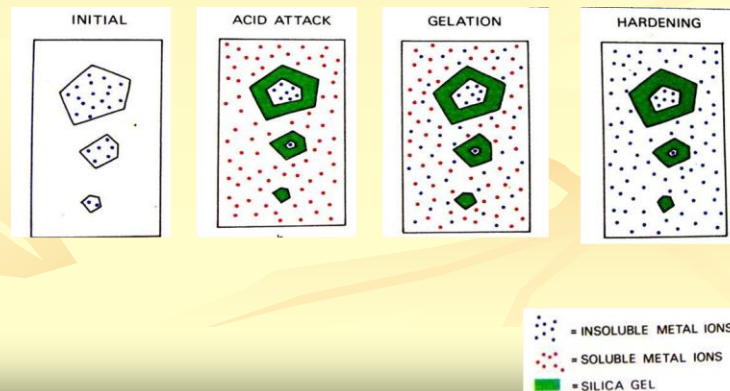
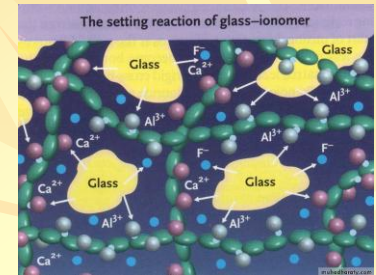
Setting Reaction

- The setting reaction of glass Ionomer cements is an acid base reaction between polyacrylic acid and fluoroaluminosilicate powder
- It has three overlapping stages..

Stage1: Dissolution

Stage2: Precipitation of salt and gelation

Stage3: Hardening and slow maturation



Mode of supply

- Powder liquid
- Pepproportioned capsules



Manipulation of GIC

Hand mixing

Hand Mixing

- The powder and liquid are dispensed onto a mixing pad
- The powder is divided into two equal portions.
- The first portion is incorporated into the liquid with a stiff spatula before the second portion is added.
- Agate spatula is used for mixing
- The mixing time is 30 to 60 seconds.



Manipulation of GIC

Hand mixing

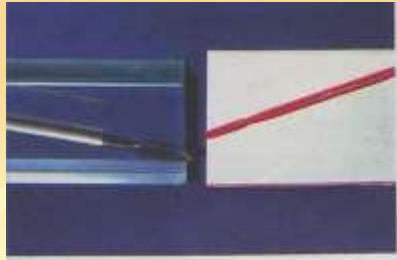


Full spoon, no excess

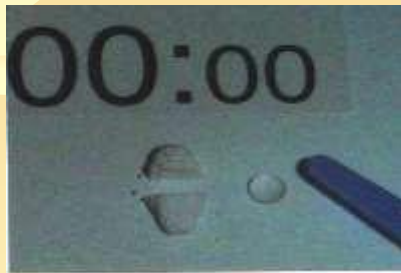


**Tip liquid bottle to side, then
invert completely
only 1 drop used.**





Liquid should not stay on paper pad longer than 1 minute (some of it may soak into it)



**Don't mix beyond 30 seconds
The objective is – only wet the particle
– no dissolving it.**



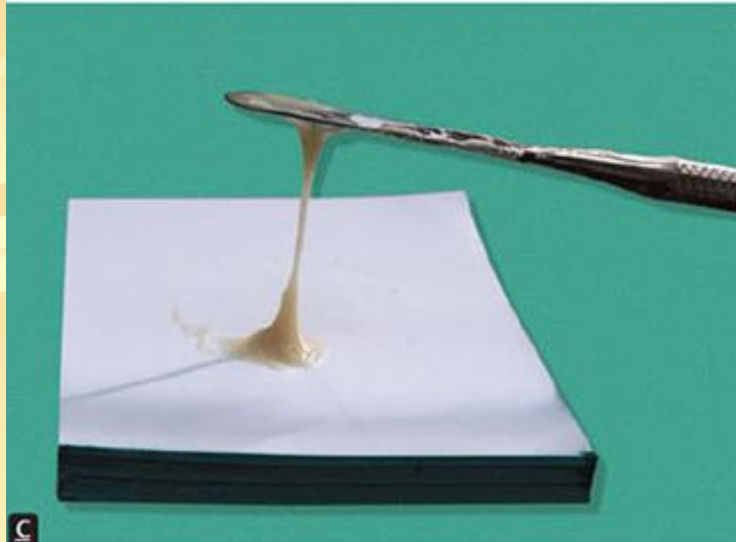
First half folded into liquid in 10-15 seconds



**Second half incorporated in 15 seconds
Small mixing area**

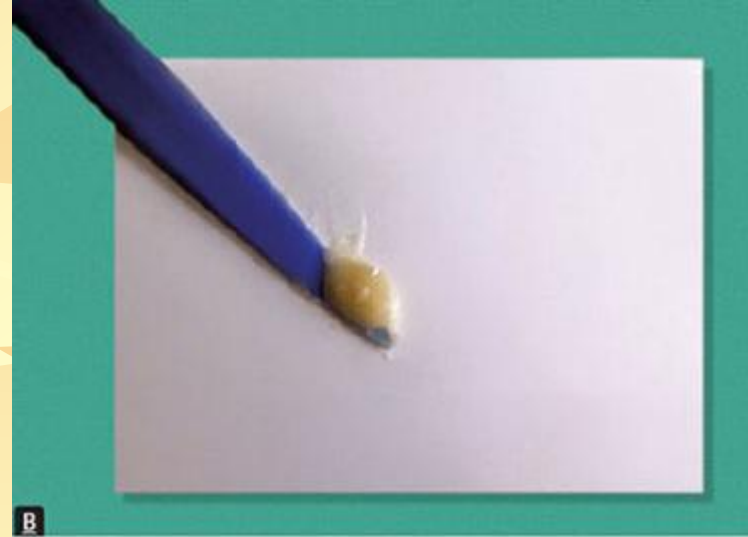
Correct consistency for hand mixed

STRING TEST



Type I : Luting : string up to 3-4cm from slab

GLOSS TEST



Type II : Glossy appearance



- Capsule GIC
- a) Capsule activation ,Before activation, shake the capsule. To activate the capsule, place the capsule into a metal GC Capsule Applier and click the lever once

Note : The capsule should be activated just before mixing and used immediately.





- b) Immediately remove the capsule and set it into a mixer (an amalgamator) and mix for 8 to 10 seconds
- c) Immediately remove the mixed capsule from the mixer and load it into the GC Capsule Gun.
- Extrude mixed cement directly into preparation. Avoid air bubbles.



Properties

1. Mechanical properties:

A) **Compressive strength:** 100 - 150 Mpa, Resin modified GIC have comparatively higher strength

B) **Tensile Strength:** 6.6 Mpa

2) Film thickness: 25 micron

3) ***Adhesion:*** Glass ionomer materials have good chemical adhesion to tooth structure

4) ***Thermal protection:*** GIC material has an adequate thermal insulating effect on the pulp serve as good insulation against thermal shock, particularly when used as liners and bases.

5) ***Flouride release:*** Beneficial characteristic of glass ionomers, is that these materials act as a reservoir of fluoride ions, Particularly beneficial for medium to high caries activity.

INDICATIONS :

Clinically, both conventional and resin-modified glass ionomers are used for a variety of restorative applications,

- Restoration of noncarious cervical lesions like abrasion, abfraction
- Liners and bases
- Restoration of primary teeth.
- Restoration of Class V carious lesions
- In patients with high caries index
- Luting of indirect restorations
- Atraumatic restorative technique

CONTRAINDICATIONS

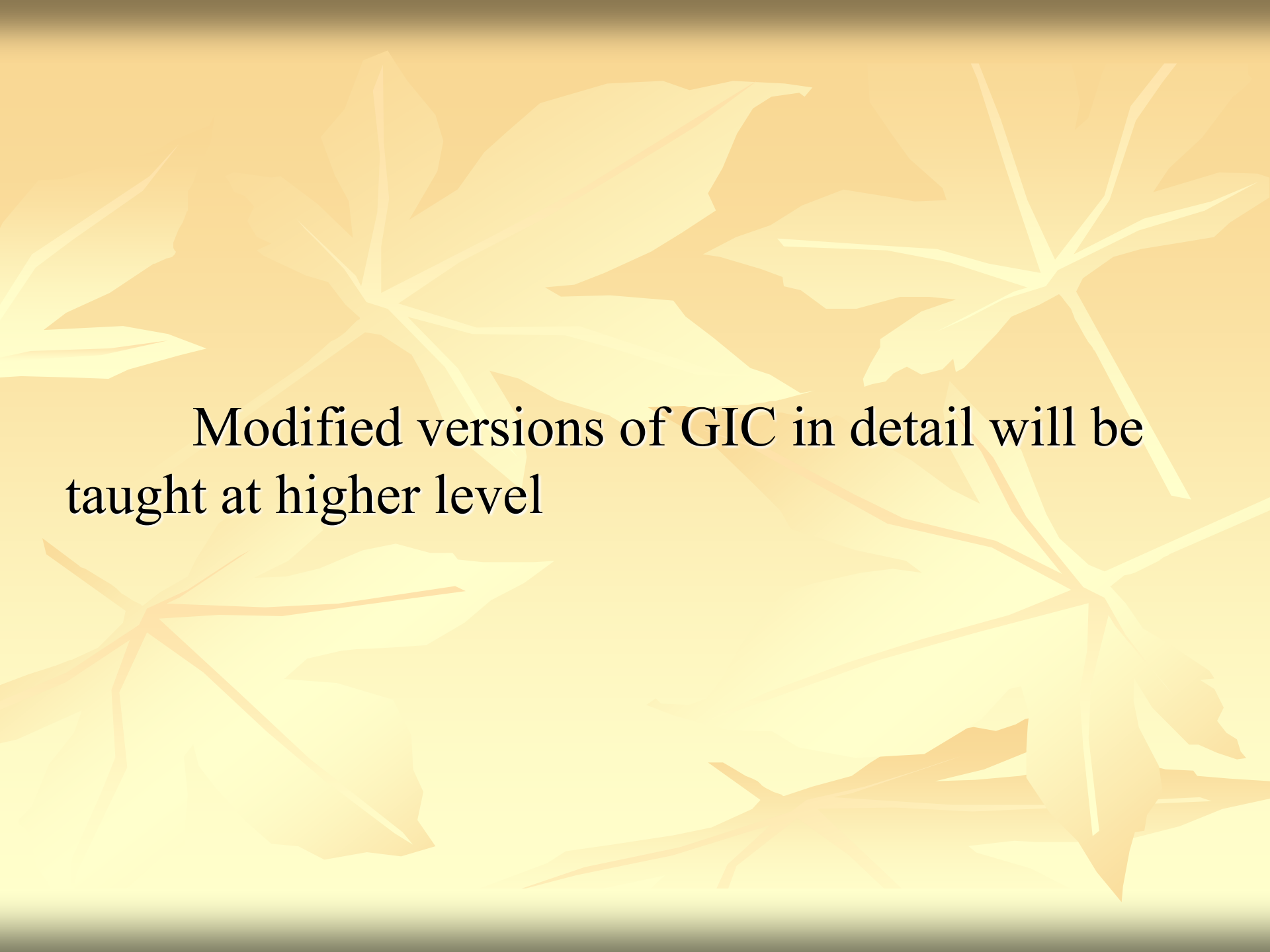
- High stress bearing areas like-Class IV fracture/caries and Conventional Class II carious lesion
- Lesions involving large areas of labial enamel where esthetics is of major importance.
- Lost cusp areas.

ADVANTAGES

- Chemical bond to the tooth.
- Anticariogenic property.
- Thermal compatibility
- Favorable pulpal response

DISADVANTAGES

- Brittle.
- Less fracture resistance, use restricted to Low stress bearing areas.
- Moisture sensitive while setting
- Less esthetics compared to composite



Modified versions of GIC in detail will be
taught at higher level

Learning Resources

- Craig & Powers, “Restorative Dental Materials” (Mosby, St. Louis). Pages: 184 - 189



Thank you